

HIGH-SPEED Li-Fi DATA TRANSFER

Using Laser Communication

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A Li-Fi (light fidelity)-based optical communication system is designed to transmit digital data as light signals using a laser diode and a photodiode. Unlike conventional RF communication systems, the design employs on-off keying (OOK) modulation, in which the presence of light represents a logic high (1) and the absence of light represents a logic low (0). The transmitted information is organised into data packets containing headers, payload, and checksum fields, enabling the reliable transfer of various file types over a free-space optical link at baud rates of up to 260kbps.

This system demonstrates high-speed optical wireless communication with serial data transmission and signal conditioning, using a laser diode, a BPW34 photodiode, and an LM393 comparator. It also highlights key communication concepts, including synchronisation, noise reduction, high-speed switching, and error detection in practical communication systems.

The technique can be adapted for smartphones and other portable devices, enabling high-speed file transfer using the built-in camera flash LEDs and optical sensors, at speeds significantly higher than many traditional wireless methods. Fig. 1 shows the author's breadboard prototype of the transmitter (L) and receiver (R) units.

Circuit and working

The system comprises two main sections: a Li-Fi laser transmitter circuit and a Li-Fi optical receiver circuit.

Fig. 2 shows the circuit diagram of the Li-Fi laser transmitter. It is

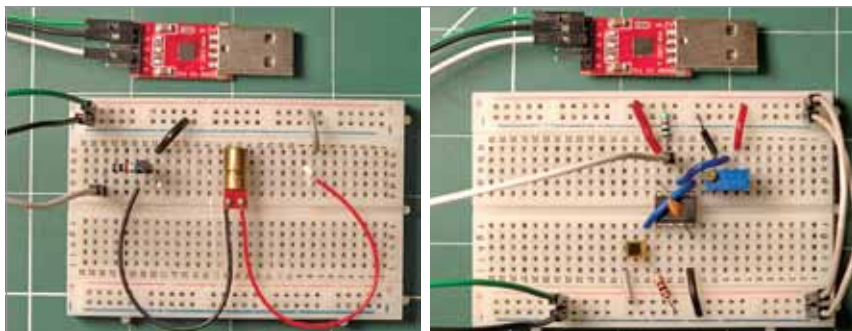


Fig. 1: Author's prototype of the transmitter (L) and receiver (R) units

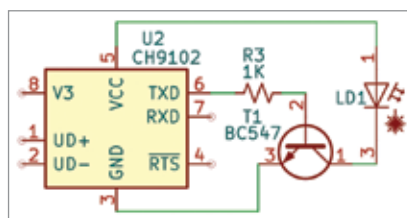


Fig. 2: Transmitter circuit

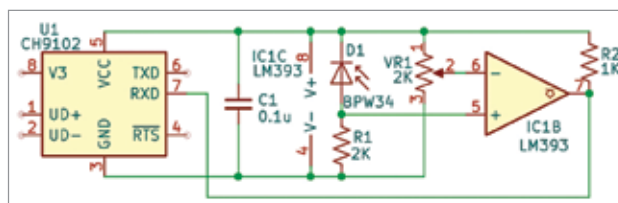


Fig. 3: Receiver circuit

built around a CH9102 USB-to-UART converter module (U2), BC547 NPN transistor (T1), and a laser diode (LD1). Both power and serial data are obtained directly from the USB interface through the CH9102 module, making the circuit suitable for direct connection to a PC or laptop USB port.

The TXD pin of the CH9102 module is connected to the base of transistor T1 through current-limiting resistor R3. The emitter of T1 is grounded, while its collector drives the laser diode connected to the supply line. During operation, the high-speed UART data from the TXD

pin switches the transistor rapidly on and off, causing the laser diode to generate corresponding optical pulses.

Due to the fast switching capability of the BC547 transistor and laser diode, the circuit is suitable for high-speed Li-Fi data transmission applications.

Fig. 3 shows the circuit diagram of the Li-Fi optical receiver. It is built around a BPW34 photodiode (D1), LM393 comparator (IC1), CH9102 USB-to-UART converter

module (U1), and a few passive components. The circuit detects the incoming laser beam and converts the optical signals back into digital data.

The BPW34 photodiode is operated in reverse-bias mode through resistor R1 to improve sensitivity and response speed. When the modulated laser light falls on the photodiode, a small voltage signal is developed across R1 and applied to the non-inverting (+) input pin 5 of the IC1B comparator. The inverting (-) input pin 6 receives an adjustable reference voltage from preset VR1, allowing sensitivity adjustment according to ambient light conditions. The IC1A